INCH-POUND
MIL-F-24402/1(SH)
3 April 1990

MILITARY SPECIFICATION SHEET

FILTER HOUSINGS, HYDRAULIC, SINGLE ELEMENT

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-F-24402.

CLASSIFICATION. Filter housings covered by this specification sheet shall be designated using part or identifying numbers (PIN) as follows:

M	24402	/1	N	B	-WXYZ
Prefix to indicate an item defined by inch-pound units	Specification number	Applicable specification sheet	Type	Element size	Indicator form

Examples: M24402/1RB-M90 is a single element housing with bypass relief designed for a size B element and equipped with a mechanical pop-up indicator which actuates at 90 lb/in^2 differential.

M24402/1NC is a non-fatigue rated, single element housing without bypass relief designed for a size C element and provided without a differential pressure indicator.

FILTER ASSEMBLY TYPE. The filter assembly type shall be designated by one of the following letters to indicate whether or not a bypass relief is installed:

N - No bypass relief
R - Bypass relief installed

ELEMENT SIZE. The element configuration and size (see the applicable specification sheet) shall be indicated by one of the following letters:

B - Size B
C - Size C

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DIFFERENTIAL PRESSURE INDICATOR FORM. This dash number suffix is included in the part number when the filter assembly is to be acquired or used with a specific type of differential pressure indicator. See MIL-F-24402/5 for indicator dash numbers. For logistic support purposes, assemblies will be acquired and stocked without differential pressure indicators. In such cases, indicators will be ordered separately using the part number shown on MIL-F-24402/5.

PART NUMBER CROSS REFERENCE. Table I provides a cross-reference between old (MIL-F-24402D and earlier) and new part numbers.

TABLE I. Cross reference of part numbers.

New part numbers		Old part numbers	
(Fatigue rated) 1/	(Non-fatigue rated) 1/	(Fatigue rated) 2/	(Non-fatigue rated) 2/
M24402/1NB	-	M24402-1N-BP-3	M24402-1N-BP-0
-	M24402-1NC	M24402-1N-CP-3	M24402-1N-CP-0
M24402/1RB	-	M24402-1R-BP-3	M24402-1R-BP-0
-	M24402-1RC	M24402-1R-CP-3	M24402-1R-CP-0

1/ If a differential pressure indicator is to be acquired with the filter assembly, add the appropriate indicator dash number suffix from the applicable specification sheet.

2/ Add E, G, or M depending on the form of differential pressure indicator acquired with the filter assembly, as follows:

E - Electrical with mechanical pop-up indicator
G - Gauge type indicator
M - Mechanical pop-up indicator

SUBSTITUTION OF ASSEMBLIES. Assemblies with new fatigue rated part numbers may be substituted for all assemblies with old part numbers as shown in table I. Assemblies with new non-fatigue rated part numbers may only be substituted for assemblies with non-fatigue rated part numbers as shown in table I. Only assemblies with old fatigue rated parts numbers may be substituted for assemblies with new fatigue rated part numbers. Assemblies with old fatigue rated and non-fatigue rated part numbers may be substituted for assemblies with new non-fatigue rated part numbers as shown in table I.

REQUIREMENTS

(a) Configuration. The filter assemblies shall meet the dimensional requirements shown on figure 1 for filter housings with size B elements installed and figure 2 for filter housings with size C elements installed.

(b) Maximum rated operating pressure. The filter assemblies shall have a maximum rated operating pressure of 3000 lb/in^2.

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(c) Rated flow capacity. The filter assemblies with size B elements installed shall have a rated flow capacity of 50 gallons per minute. The filter assemblies with size C elements installed shall have a rated flow capacity of 25 gallons per minute.

(d) Performance.

(1) Bypass relief requirements. Type R assemblies shall have bypass relief valve settings as specified in table II.

TABLE II. Bypass relief valve setting.

Test condition	Pressure (lb/in^2)	Test fluid temperature	Leakage rate (mL/min)	
			MIL-L-17331	MIL-H-5606
(a) Cracking pressure				
(1) Qualification	150 maximum	40°C maximum	10 minimum	
(2) Quality conformance	Not more than 5 lb/in^2 higher than the actual cracking pressure for the qualification assembly.	20 to 60°C	10 minimum	50 minimum
(b) Full flow (qualification only)	200 maximum	40°C maximum	N/A	
(c) Reseat pressure				
(1) Qualification	120 minimum	40°C minimum	2 maximum	
(2) Quality conformance	120 minimum	20 to 60°C	2 maximum	4 maximum

(2) Fatigue impulse rating. The filter assemblies with size B elements shall have a fatigue impulse rating of 3000 lb/in^2 in accordance with NFPA T2.6.1.

(3) Proof pressure. The filter assemblies shall withstand a proof pressure of 4500 lb/in^2.

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(4) Housing pressure drop. At the maximum rated flow with MIL-L-17331 fluid, the housing pressure drop with a minimum restriction dummy element installed shall be not greater than 25 lb/in^2 differential.

(5) Burst pressure. The non-fatigue rated filter assemblies with size C elements shall withstand a burst pressure of 7500 lb/in^2.

Preparing activity:
Navy - SH
(Project 4330-N057-01)

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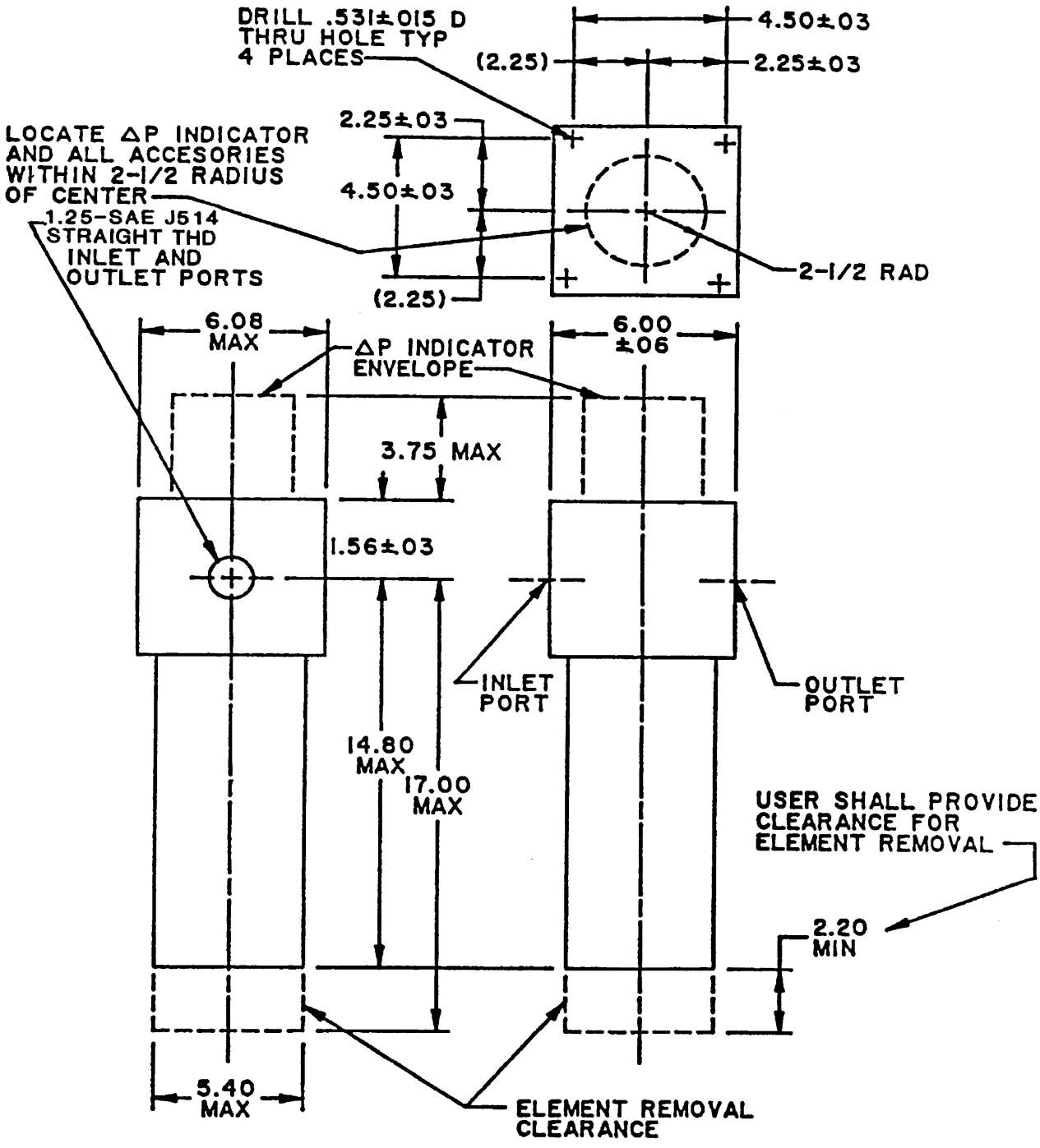


FIGURE 1. Single element (size B) filter housing outline dimensions.

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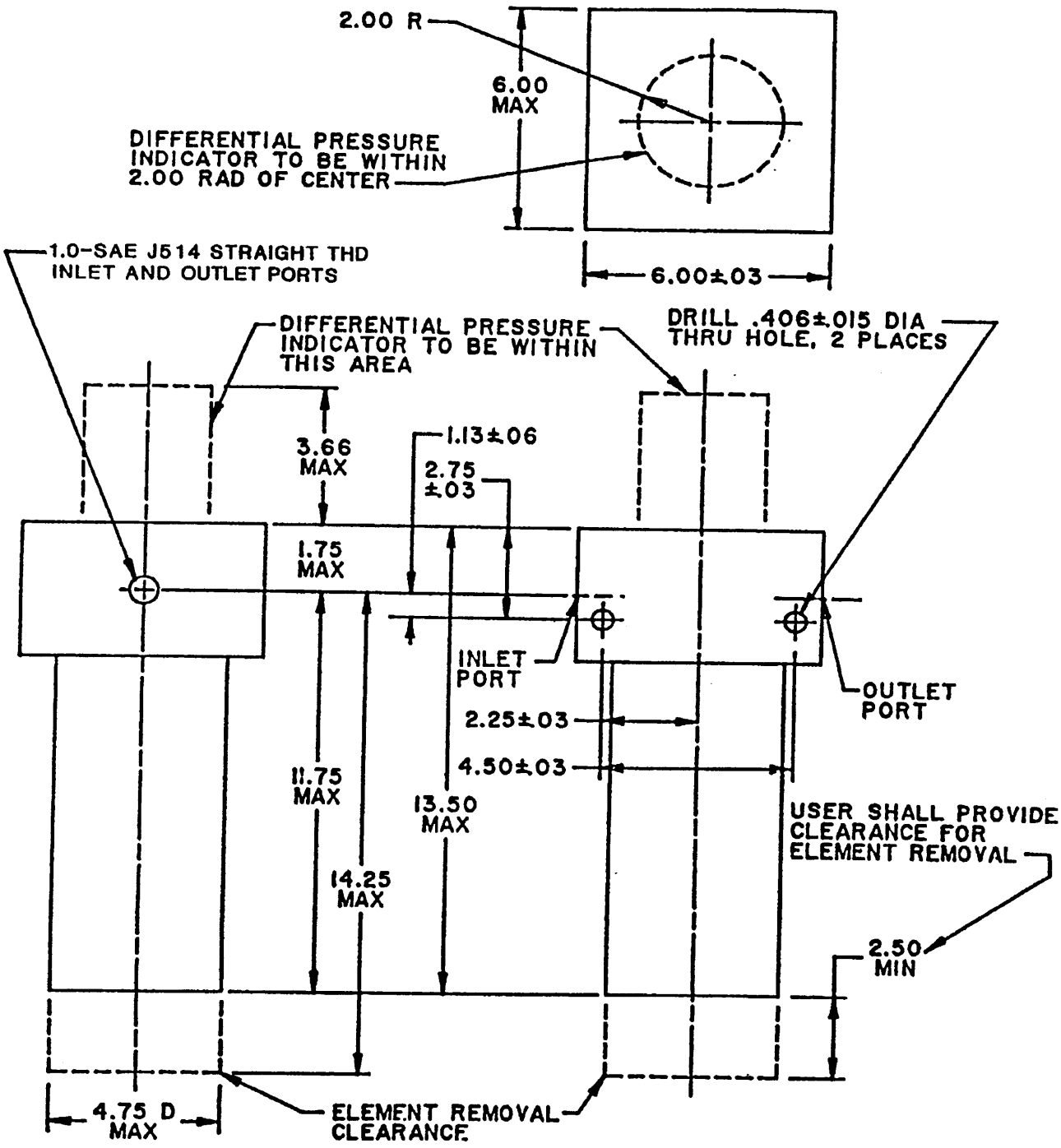


FIGURE 2. Single element (size C) filter housing outline dimensions.

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